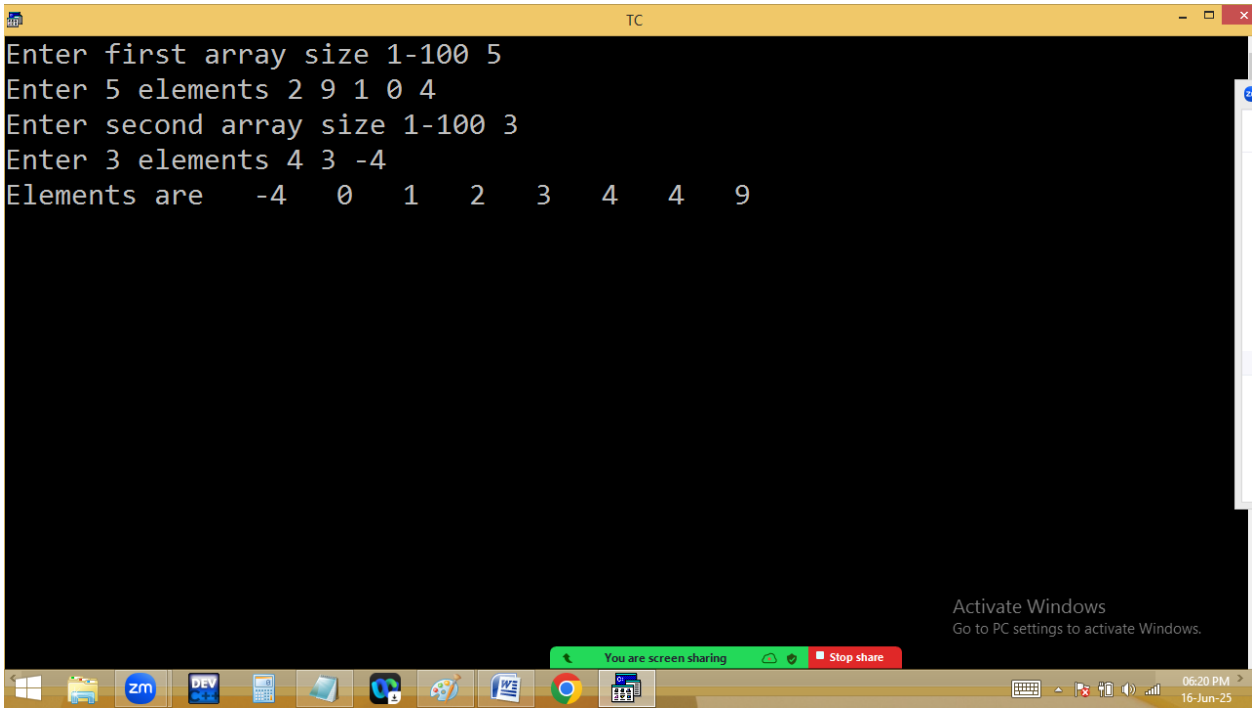
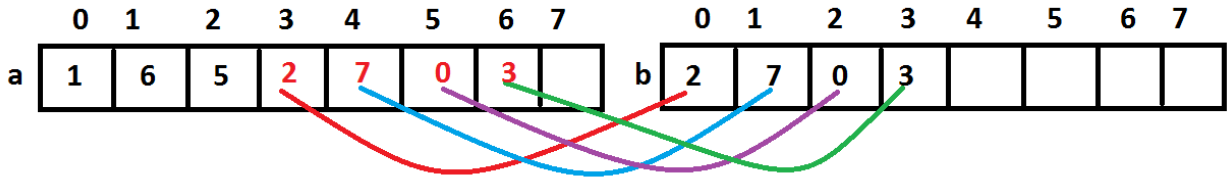
Merging of arrays:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],b[100],s1,s2,i,j;
clrscr();
printf("Enter first array size 1-100 ");scanf("%d",&s1);
printf("Enter %d elements ",s1);
for(i=0;i<s1;i++)scanf("%d",&a[i]);
printf("Enter second array size 1-100 ");scanf("%d",&s2);
printf("Enter %d elements ",s2);
for(i=0;i<s2;i++)scanf("%d",&b[i]);
for(i=s1,j=0;j<s2;j++,i++)a[i]=b[j];
for(i=0;i<=s1+s2-2;i++)
{
for(j=i+1;j<=s1+s2-1;j++)
```

```
{
if(a[i]>a[j])
{
int t=a[i];a[i]=a[j];a[j]=t;
}
}
}
printf("Elements are
");for(i=0;i<s1+s2;i++)printf("%4d",a[i]);
getch();
}
```





for(i=3,j=0;j<4;j++,i++)a[i]=b[j];

3 = 0
4 = 1
5 = 2
6 = 3

2-d arrays:

Array with several rows and several columns is called 2-d array.

Array with 2 subscripting operators [ ] [ ]

It is an implicit double pointer.

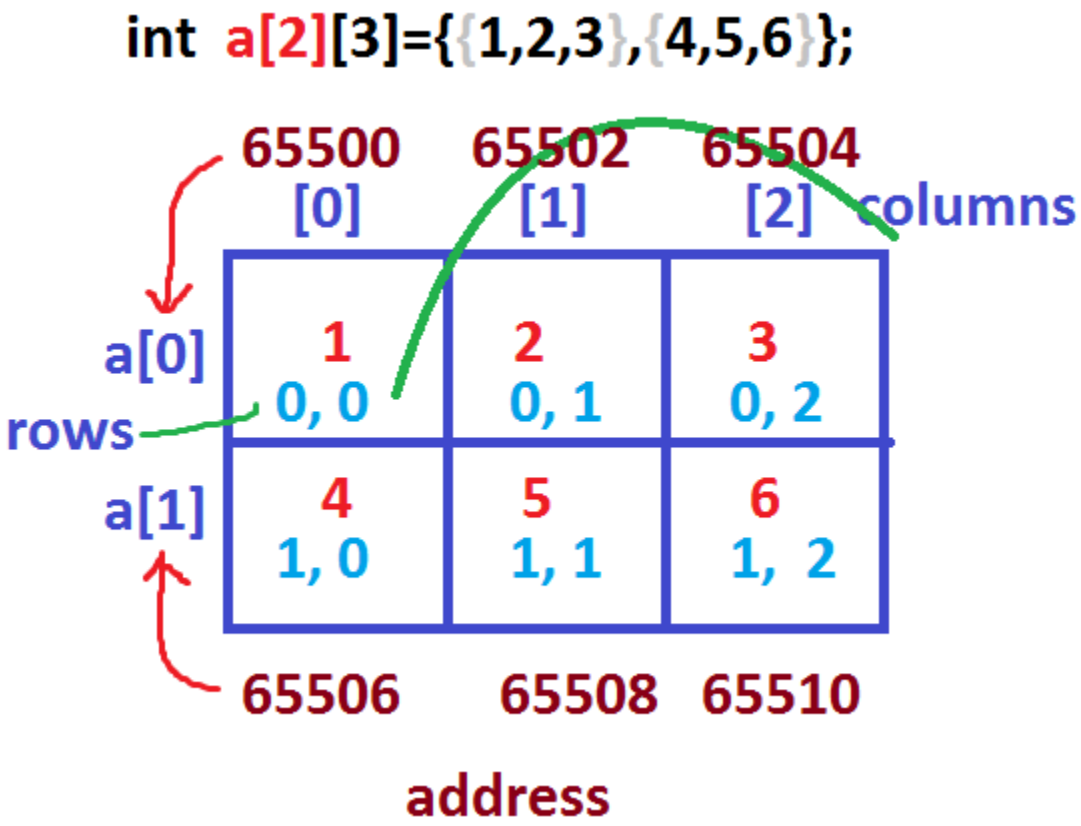
It is m*n matrix.

It is collection of one dimensional arrays. i.e. it is array of array.

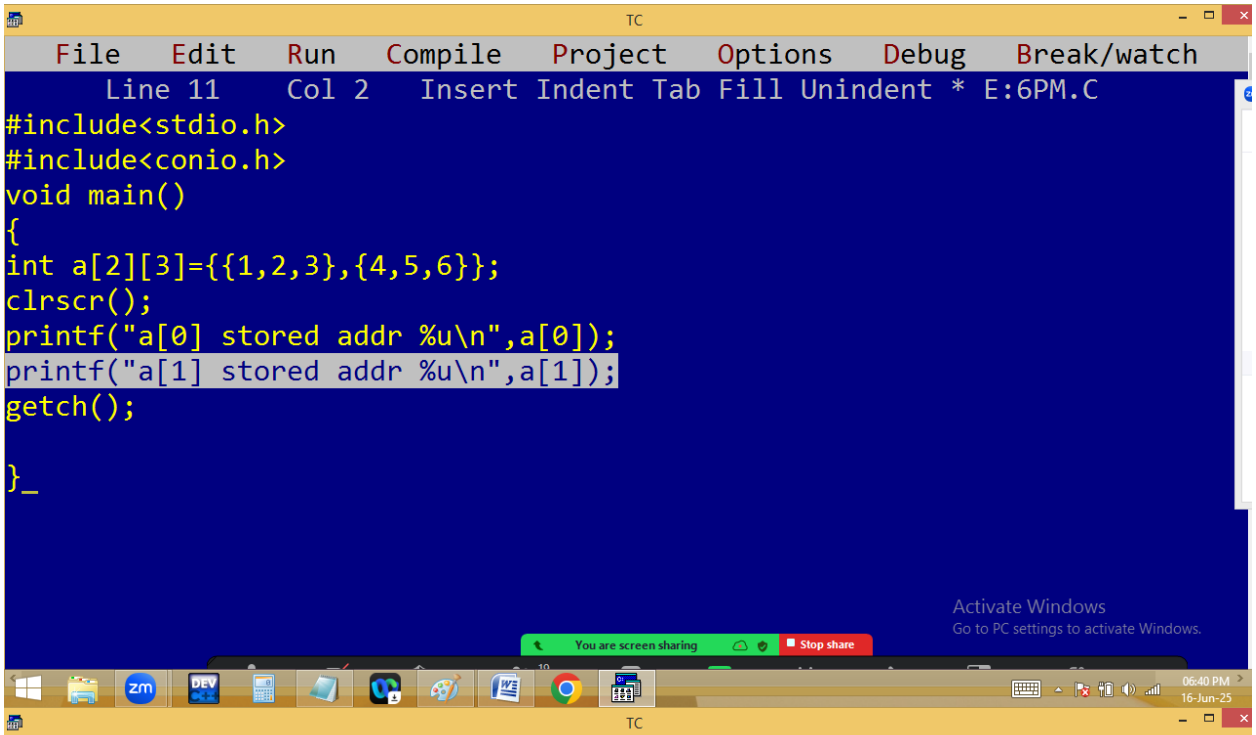
Syntax:

datatype variable [rows] [cols] = { elements };

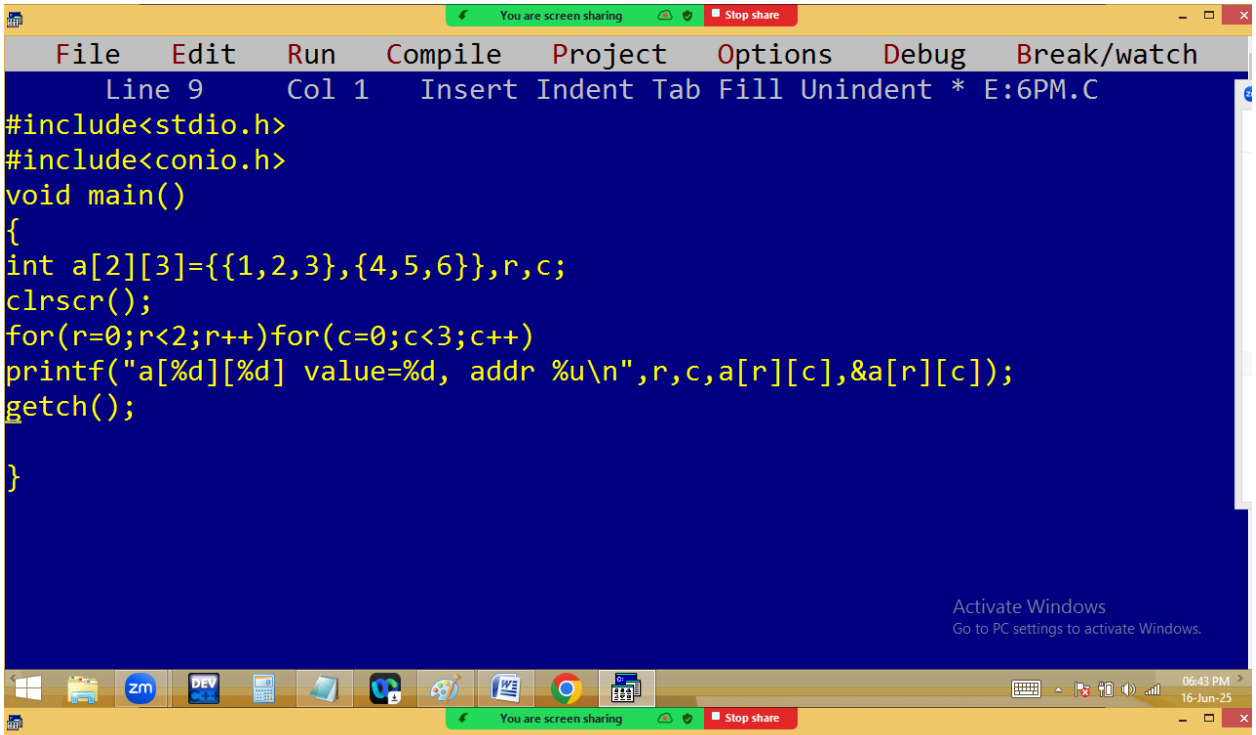
Eg: int a[2][3] = { {1,2,3},{4,5,6}};

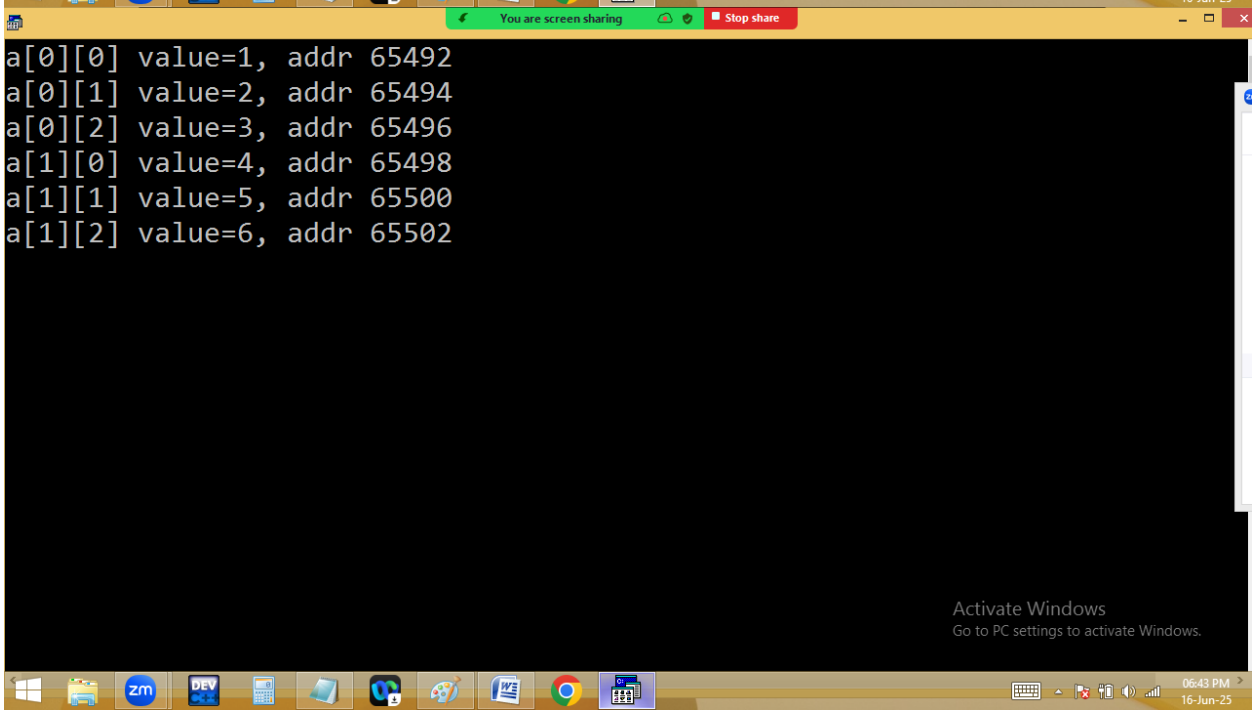


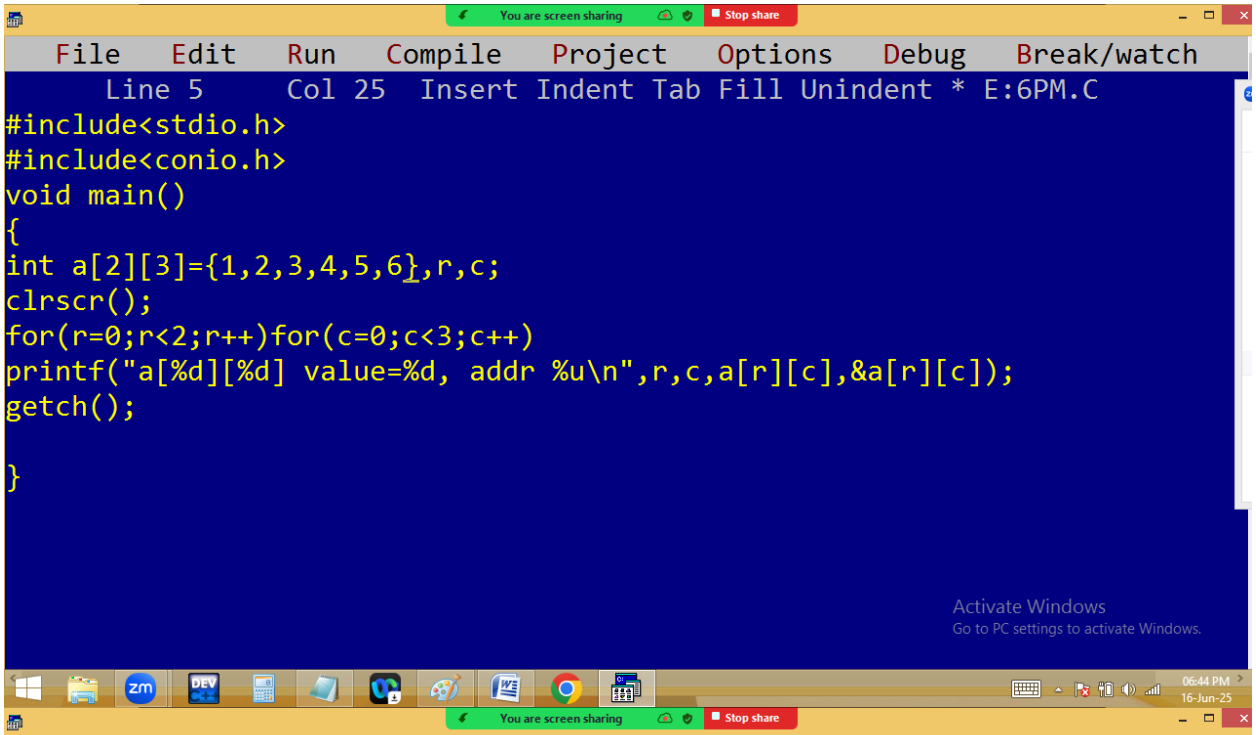
Finding address of 2-d array:

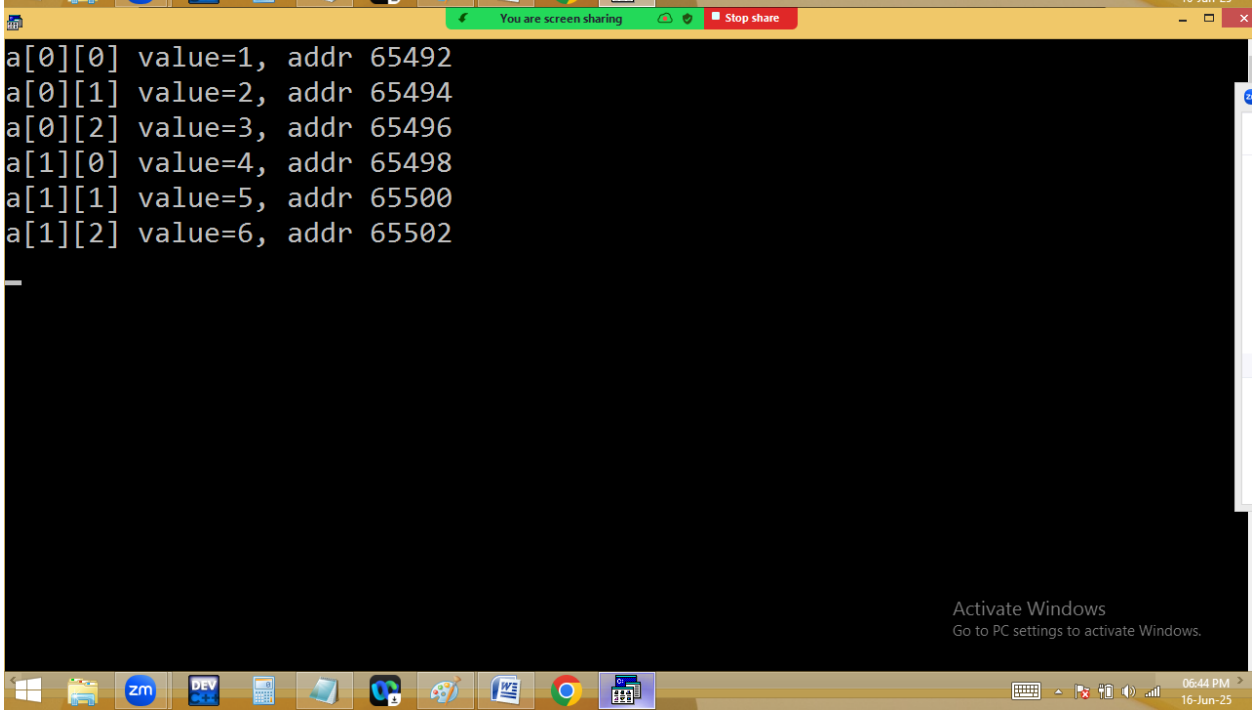

TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 2 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3},{4,5,6}};
clrscr();
printf("a[0] stored addr %u\n",a[0]);
printf("a[1] stored addr %u\n",a[1]);
getch();
}
You are screen sharing
Stop share
Activate Windows
Go to PC settings to activate Windows.
06:40 PM
16-Jun-25

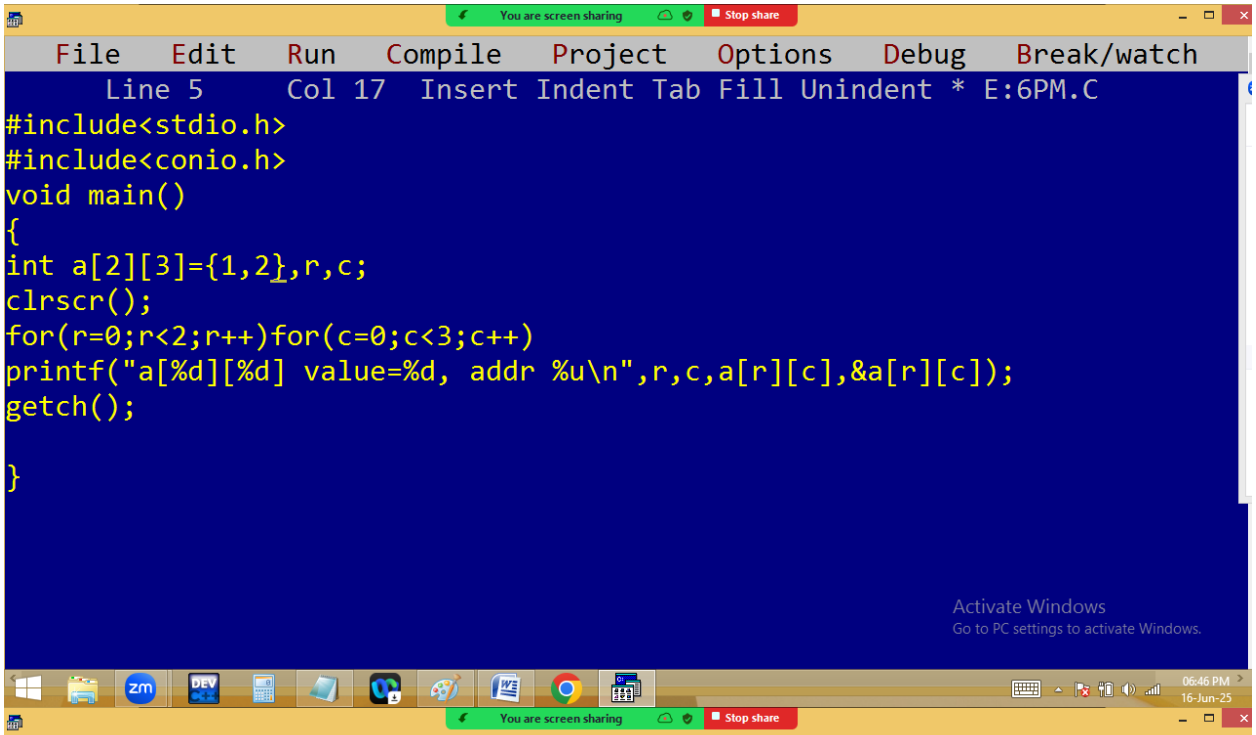

TC
a[0] stored addr 65492
a[1] stored addr 65498
You are screen sharing
Stop share
Mute
Start Video
Security
Participants
20
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
Activate Windows
06:40 PM
16-Jun-25

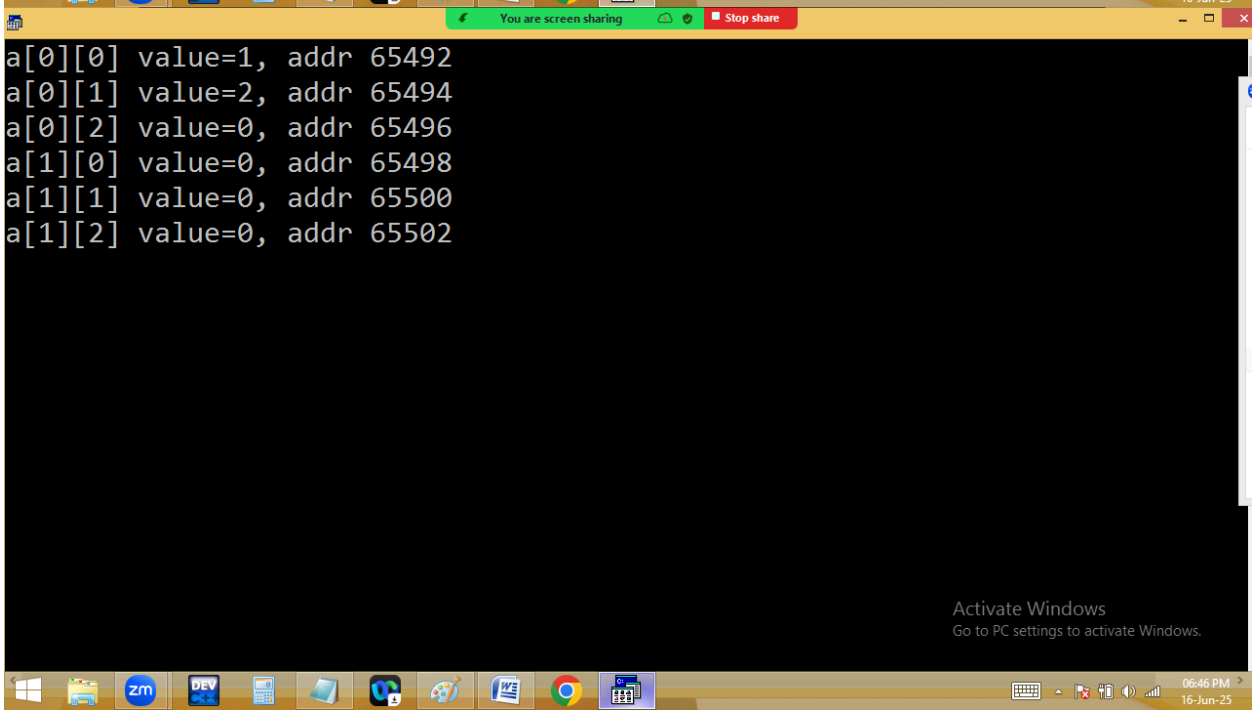

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3},{4,5,6}},r,c;
clrscr();
for(r=0;r<2;r++)for(c=0;c<3;c++)
printf("a[%d][%d] value=%d, addr %u\n",r,c,a[r][c],&a[r][c]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:43 PM
16-Jun-25

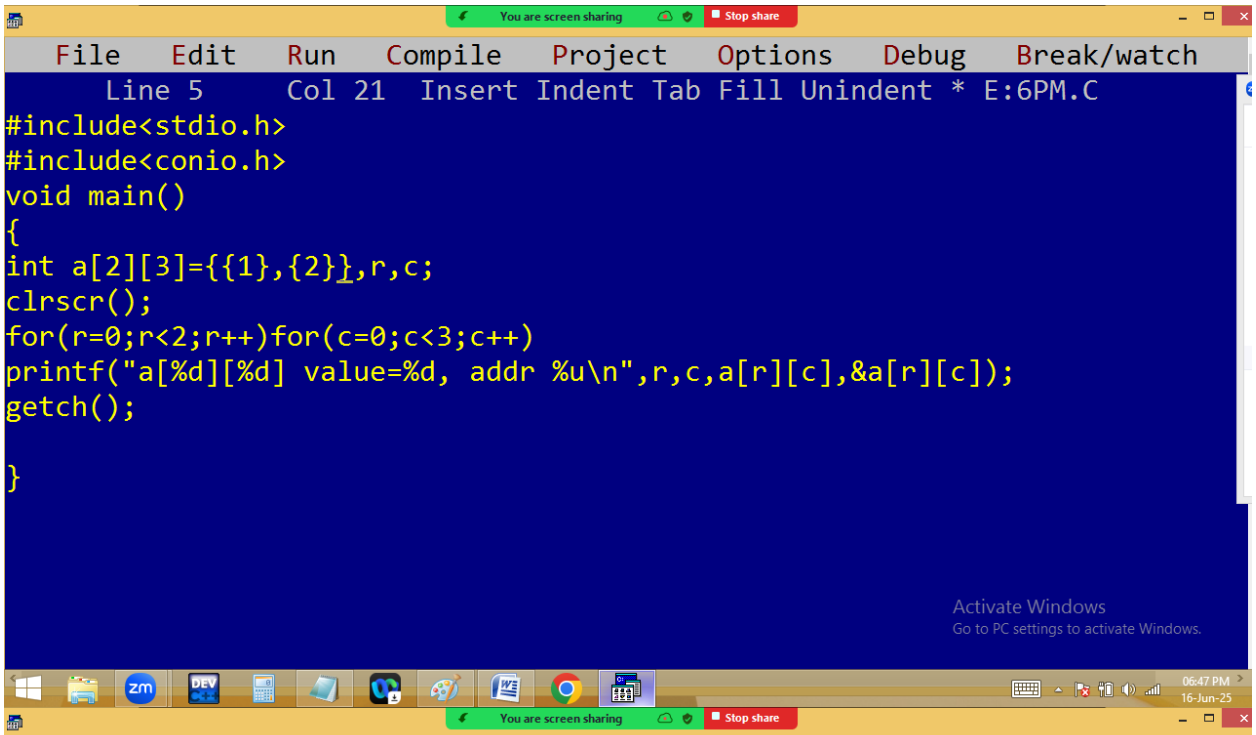

You are screen sharing
Stop share
a[0][0] value=1, addr 65492
a[0][1] value=2, addr 65494
a[0][2] value=3, addr 65496
a[1][0] value=4, addr 65498
a[1][1] value=5, addr 65500
a[1][2] value=6, addr 65502
Activate Windows
Go to PC settings to activate Windows.
06:43 PM
16-Jun-25

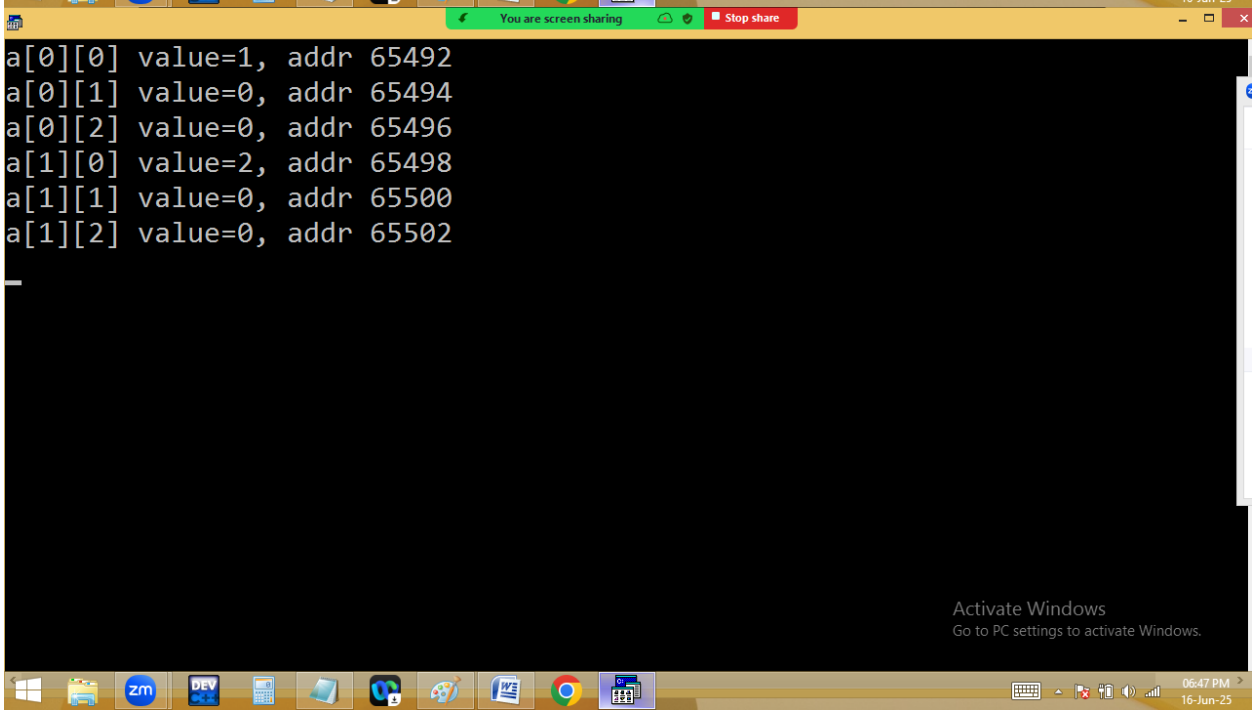

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 25 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
for(r=0;r<2;r++)for(c=0;c<3;c++)
printf("a[%d][%d] value=%d, addr %u\n",r,c,a[r][c],&a[r][c]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:44 PM
16-Jun-25

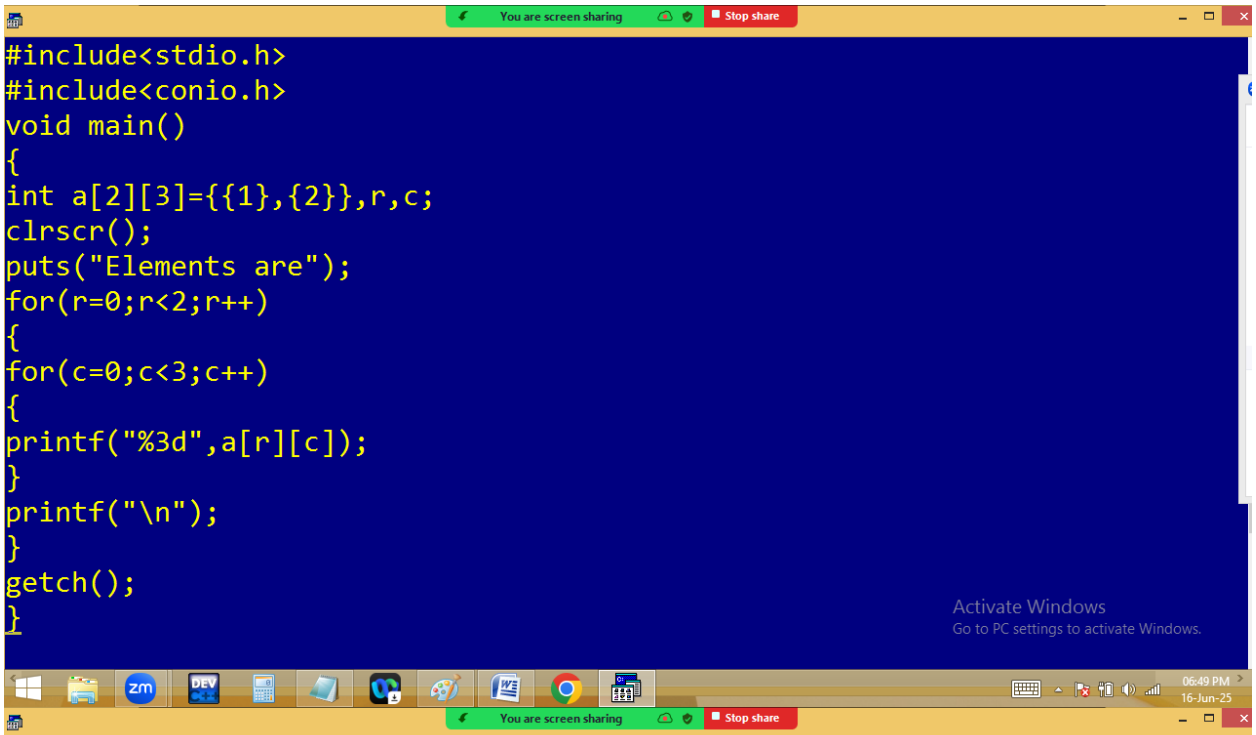

You are screen sharing
Stop share
a[0][0] value=1, addr 65492
a[0][1] value=2, addr 65494
a[0][2] value=3, addr 65496
a[1][0] value=4, addr 65498
a[1][1] value=5, addr 65500
a[1][2] value=6, addr 65502
Activate Windows
Go to PC settings to activate Windows.
06:44 PM
16-Jun-25

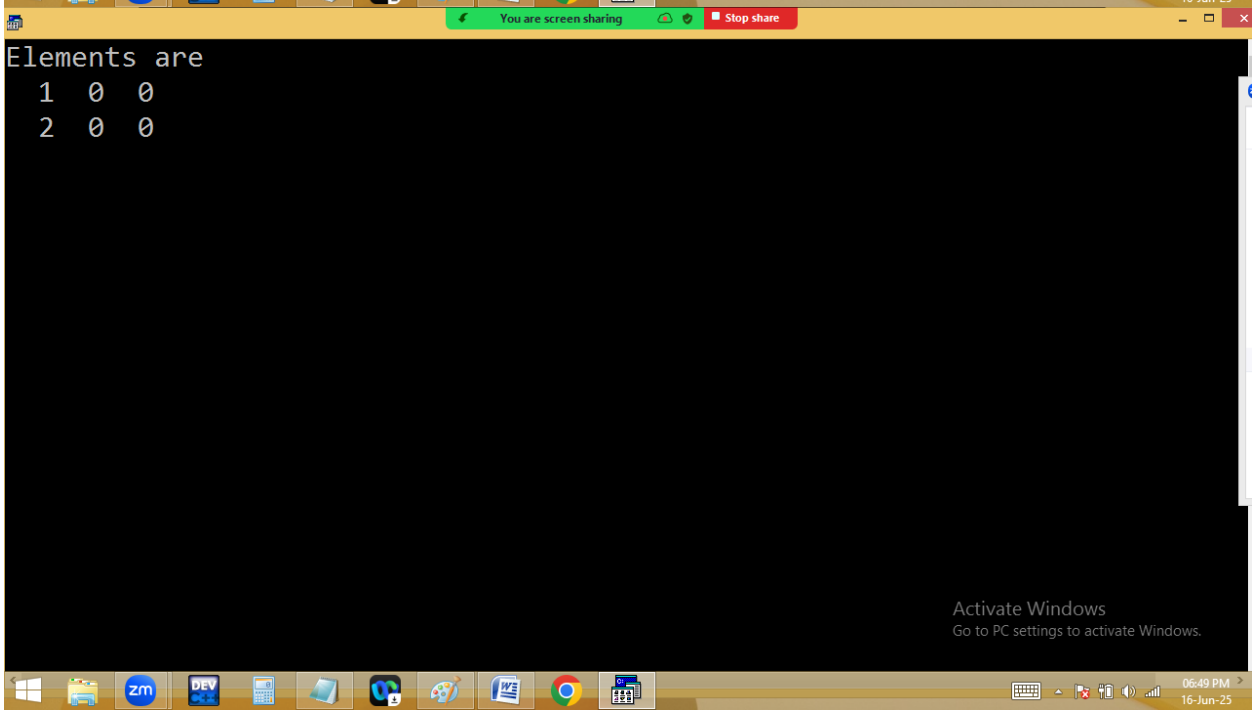

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 17 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2},r,c;
clrscr();
for(r=0;r<2;r++)for(c=0;c<3;c++)
printf("a[%d][%d] value=%d, addr %u\n",r,c,a[r][c],&a[r][c]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:46 PM
16-Jun-25

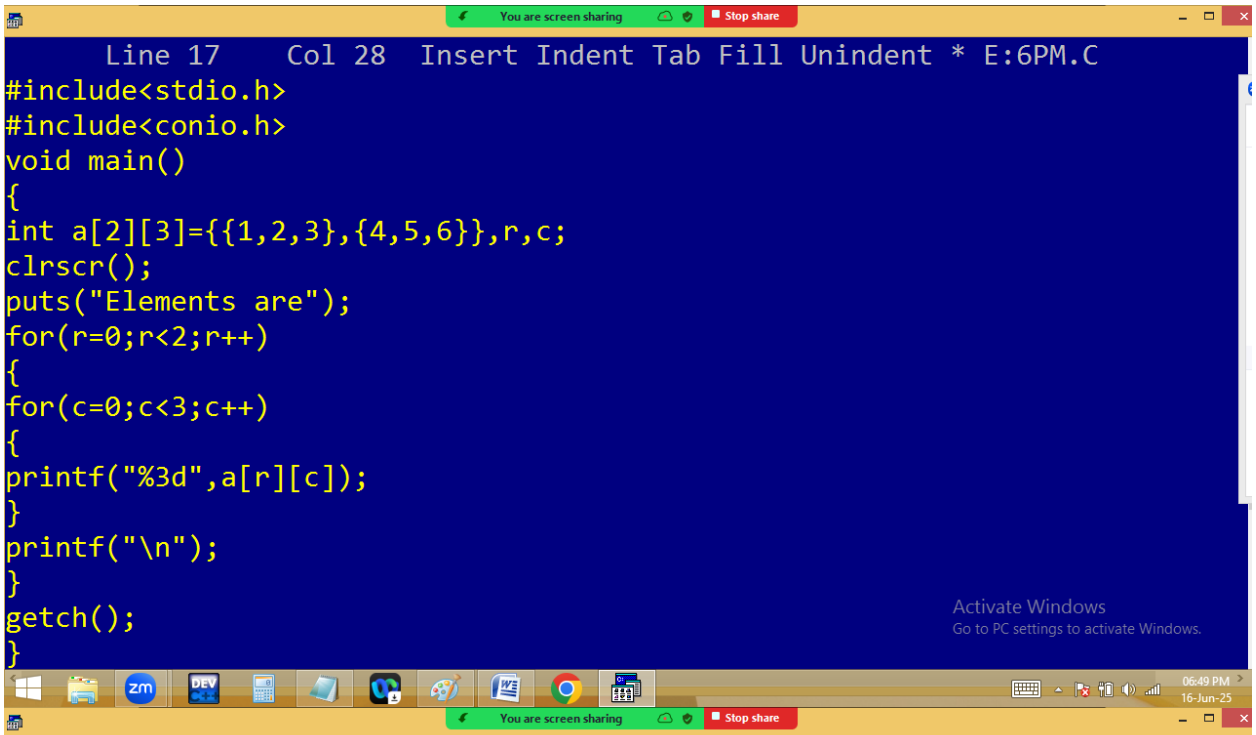

You are screen sharing
Stop share
a[0][0] value=1, addr 65492
a[0][1] value=2, addr 65494
a[0][2] value=0, addr 65496
a[1][0] value=0, addr 65498
a[1][1] value=0, addr 65500
a[1][2] value=0, addr 65502
Activate Windows
Go to PC settings to activate Windows.
06:46 PM
16-Jun-25

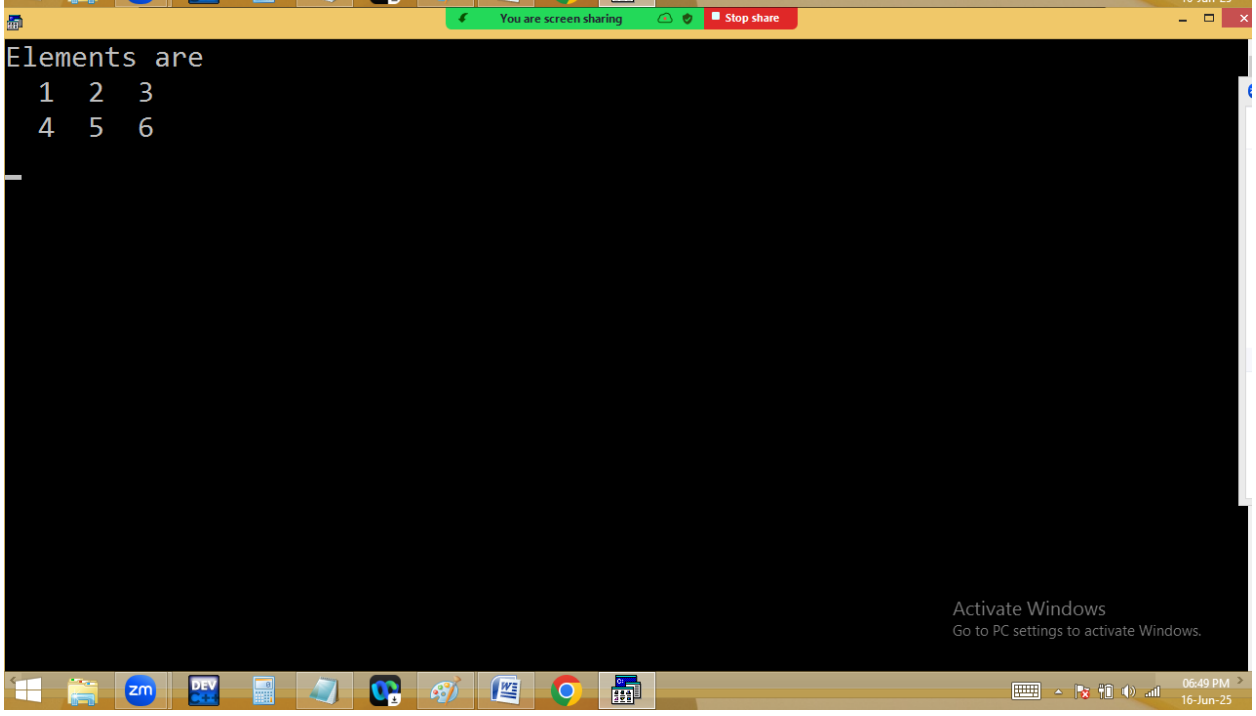

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 21 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1},{2}},r,c;
clrscr();
for(r=0;r<2;r++)for(c=0;c<3;c++)
printf("a[%d][%d] value=%d, addr %u\n",r,c,a[r][c],&a[r][c]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:47 PM
16-Jun-25

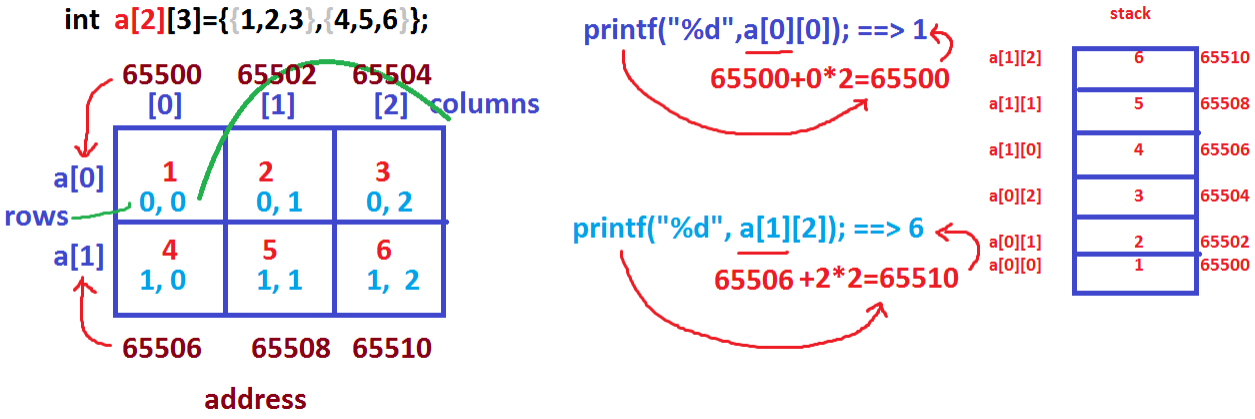

You are screen sharing
Stop share
a[0][0] value=1, addr 65492
a[0][1] value=0, addr 65494
a[0][2] value=0, addr 65496
a[1][0] value=2, addr 65498
a[1][1] value=0, addr 65500
a[1][2] value=0, addr 65502
Activate Windows
Go to PC settings to activate Windows.
06:47 PM
16-Jun-25


You are screen sharing
Stop share
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1},{2}},r,c;
clrscr();
puts("Elements are");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%3d",a[r][c]);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:49 PM
16-Jun-25

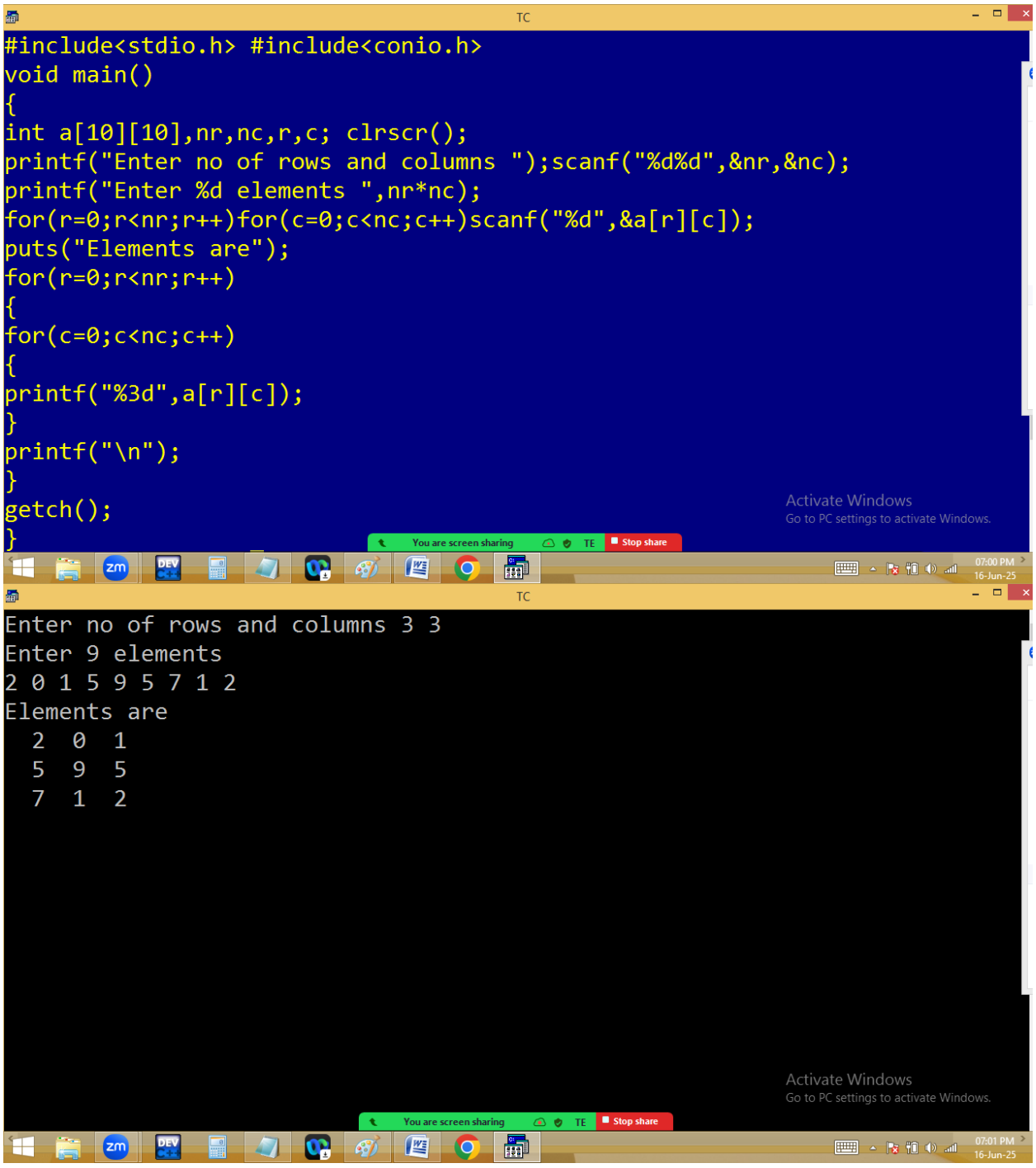

You are screen sharing
Stop share
Elements are
1 0 0
2 0 0
Activate Windows
Go to PC settings to activate Windows.
06:49 PM
16-Jun-25

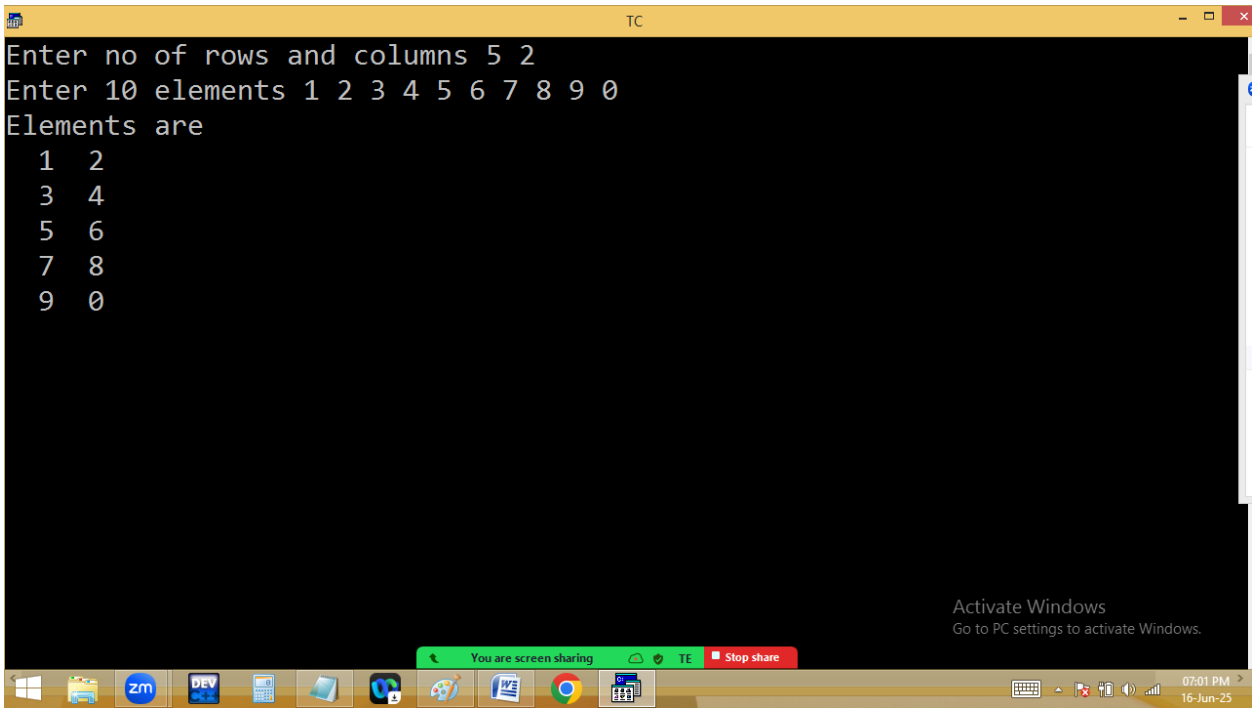

You are screen sharing
Stop share
Line 17 Col 28 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3},{4,5,6}},r,c;
clrscr();
puts("Elements are");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%3d",a[r][c]);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:49 PM
16-Jun-25


You are screen sharing
Stop share
Elements are
1 2 3
4 5 6
Activate Windows
Go to PC settings to activate Windows.
06:49 PM
16-Jun-25

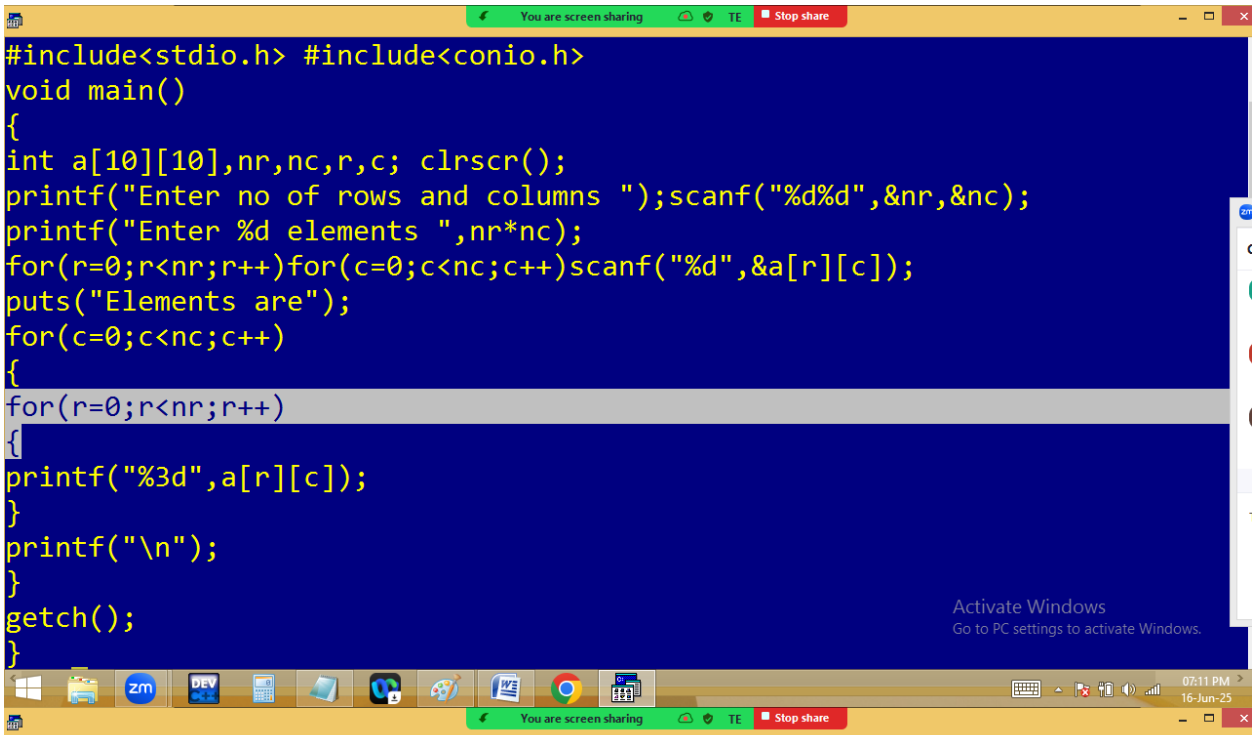


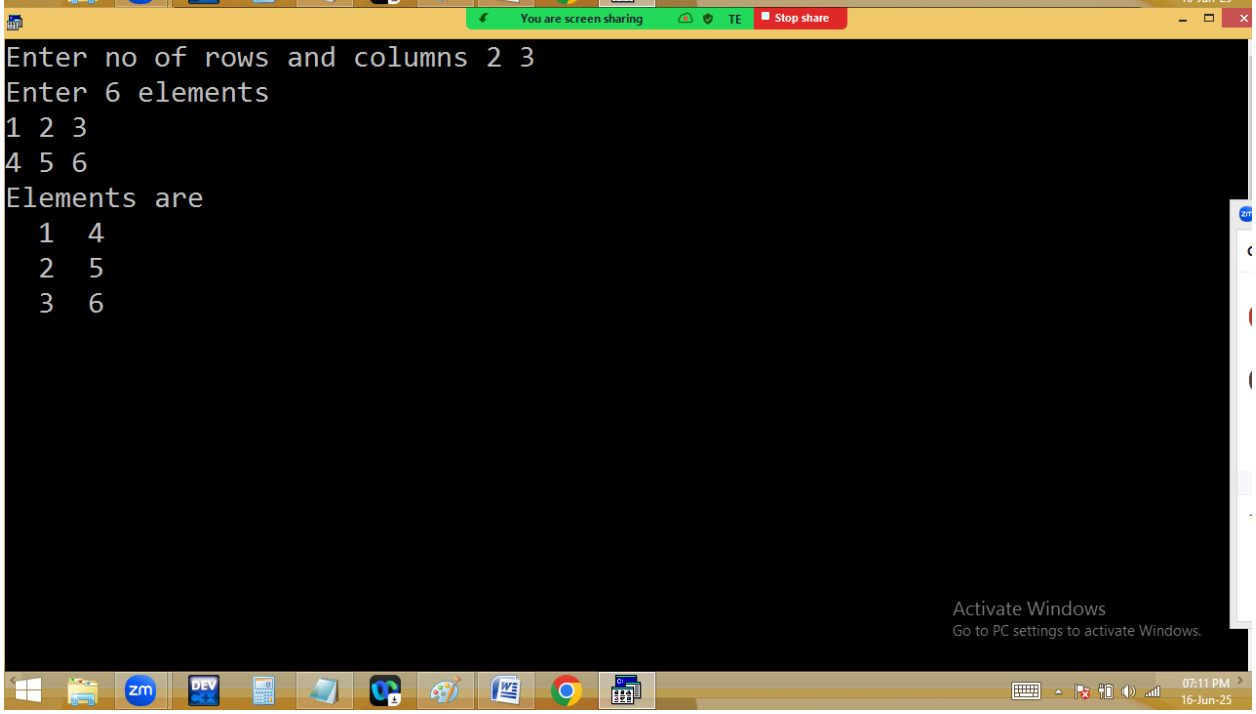
Reading and printing elements of n*n matrix:


TC
#include<stdio.h> #include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Elements are");
for(r=0;r<nr;r++)
{
for(c=0;c<nc;c++)
{
printf("%3d",a[r][c]);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
TE
Stop share
07:00 PM
16-Jun-25
TC
Enter no of rows and columns 3 3
Enter 9 elements
2 0 1 5 9 5 7 1 2
Elements are
2 0 1
5 9 5
7 1 2
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
TE
Stop share
07:01 PM
16-Jun-25



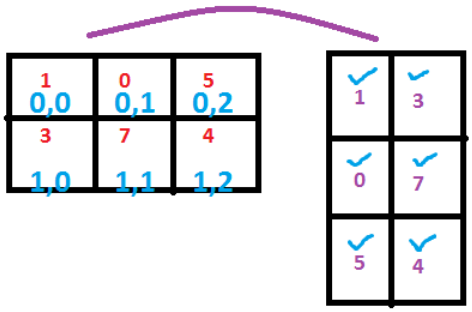
Transpose of n*n matrix:

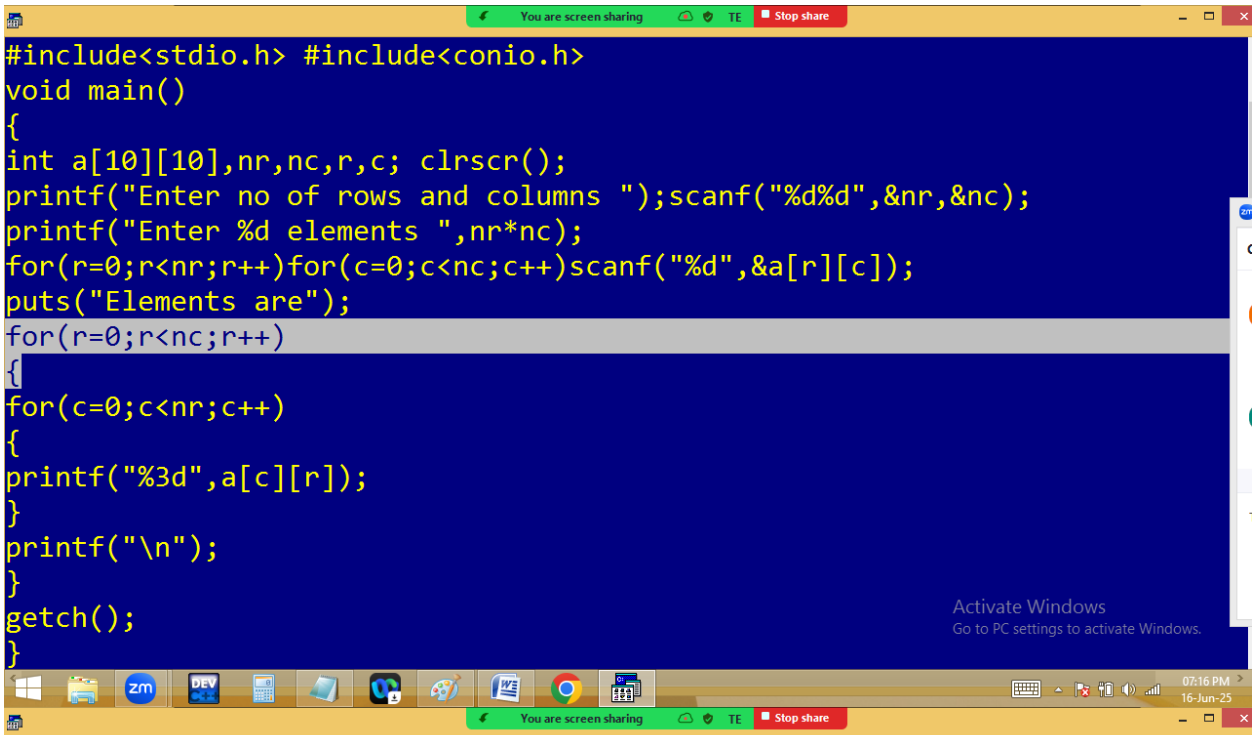

You are screen sharing
Stop share
#include<stdio.h> #include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Elements are");
for(c=0;c<nc;c++)
{
for(r=0;r<nr;r++)
{
printf("%3d",a[r][c]);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:11 PM
16-Jun-25


You are screen sharing
Stop share
Enter no of rows and columns 2 3
Enter 6 elements
1 2 3
4 5 6
Elements are
1 4
2 5
3 6
Activate Windows
Go to PC settings to activate Windows.
07:11 PM
16-Jun-25

```
for( c=0; c<3; c++)
{
for( r=0; r<2; r++)
{
p(a[r][c]);
}
p("\n");
}
```

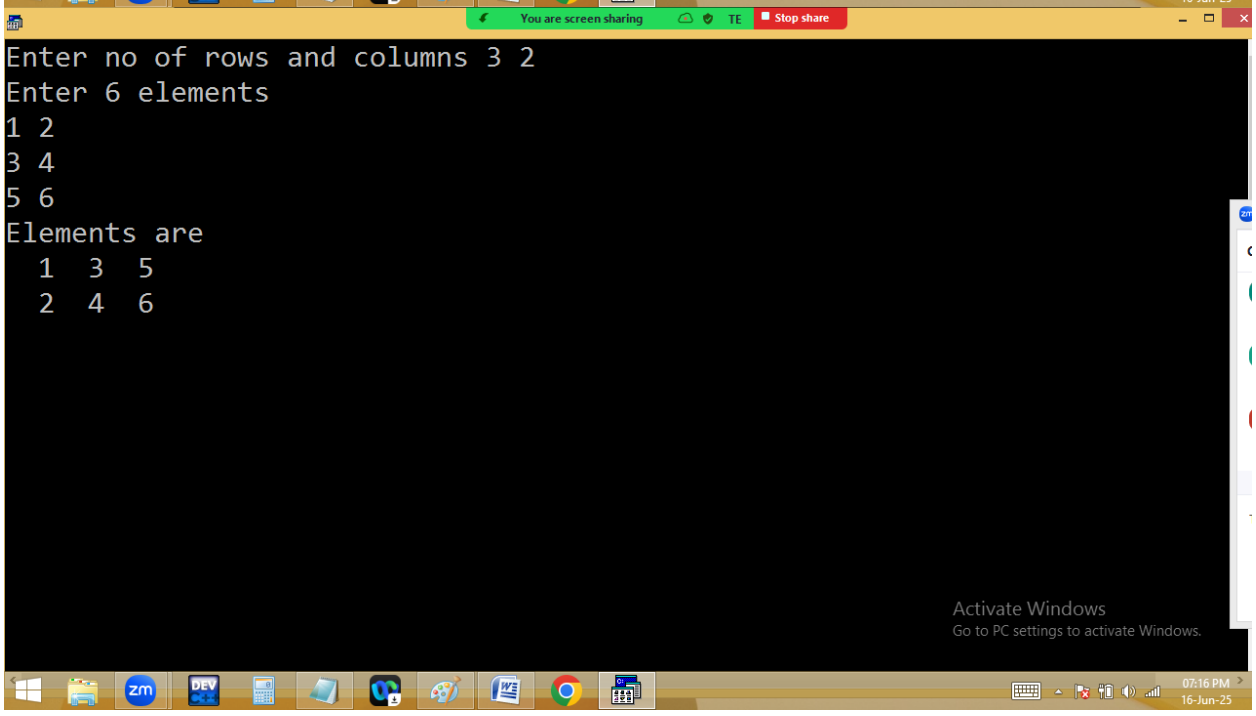
r	c
0 1	0
0 1	1
0 1	2
	3





```
#include<stdio.h> #include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Elements are");
for(r=0;r<nc;r++)
{
for(c=0;c<nr;c++)
{
printf("%3d",a[c][r]);
}
printf("\n");
}
getch();
}
```

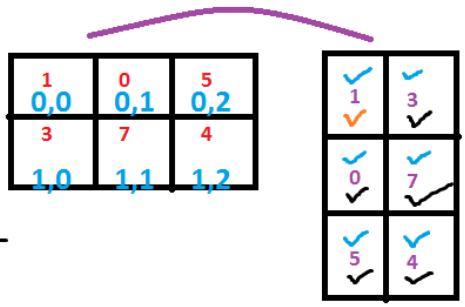
You are screen sharing TE Stop share
Activate Windows
Go to PC settings to activate Windows.
07:16 PM
16-Jun-25

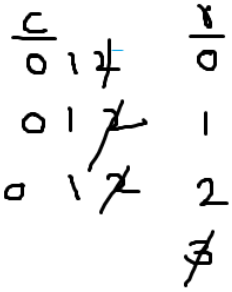


```
Enter no of rows and columns 3 2
Enter 6 elements
1 2
3 4
5 6
Elements are
  1  3  5
  2  4  6
```

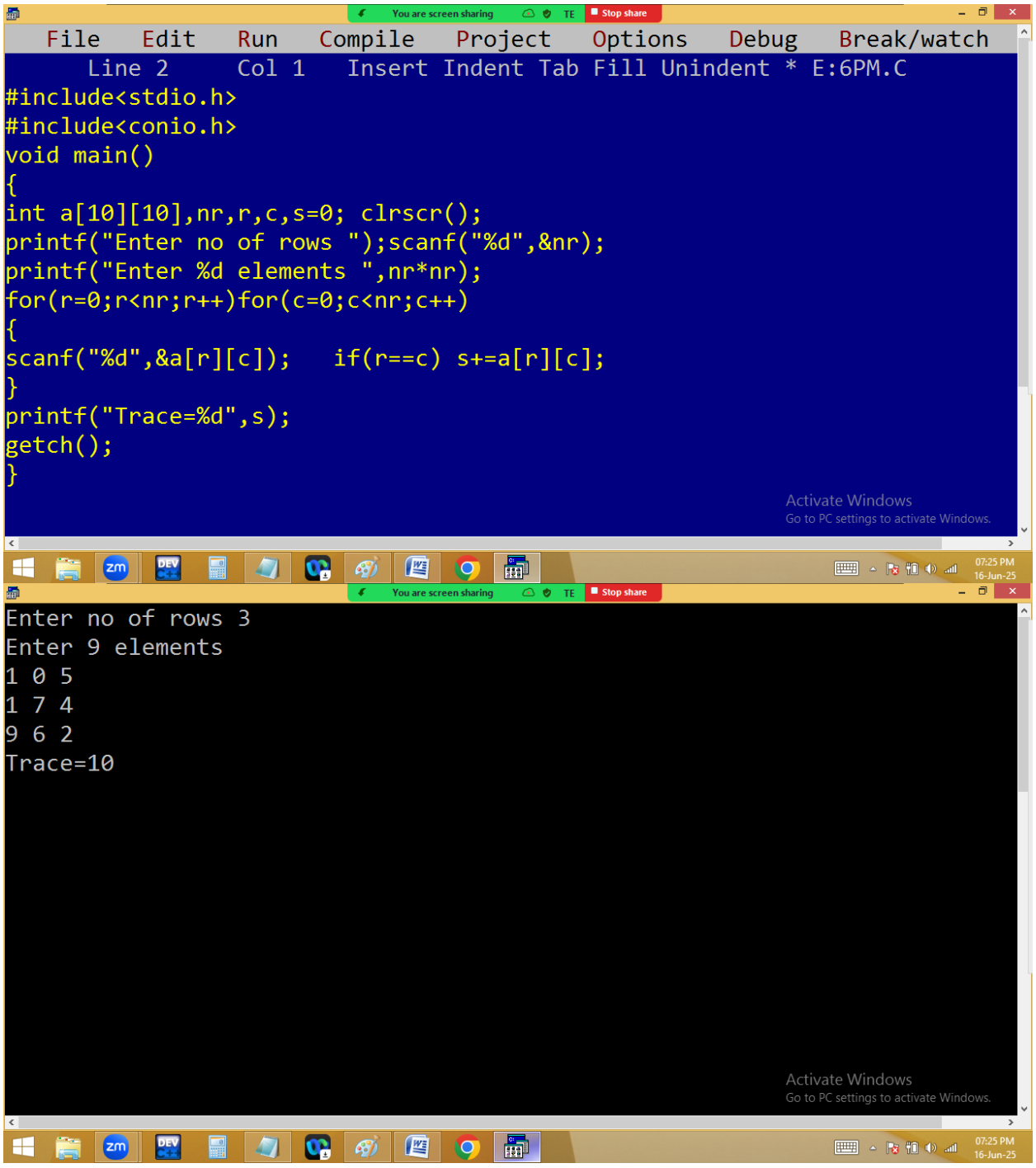
You are screen sharing TE Stop share
Activate Windows
Go to PC settings to activate Windows.
07:16 PM
16-Jun-25

```
for( r=0; r<3; r++)
{
for( c=0; c<2; c++)
{
p(a[c][r]);
}
p("\n");
}
```



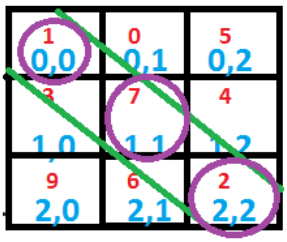


Find the trace of n*n matrix:


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 2 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,r,c,s=0; clrscr();
printf("Enter no of rows ");scanf("%d",&nr);
printf("Enter %d elements ",nr*nr);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)
{
scanf("%d",&a[r][c]); if(r==c) s+=a[r][c];
}
printf("Trace=%d",s);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:25 PM
16-Jun-25
You are screen sharing
Stop share
Enter no of rows 3
Enter 9 elements
1 0 5
1 7 4
9 6 2
Trace=10
Activate Windows
Go to PC settings to activate Windows.
07:25 PM
16-Jun-25

sum of principle diagonal elements is called trace

```
if( r==c )s+=a[r][c];
```



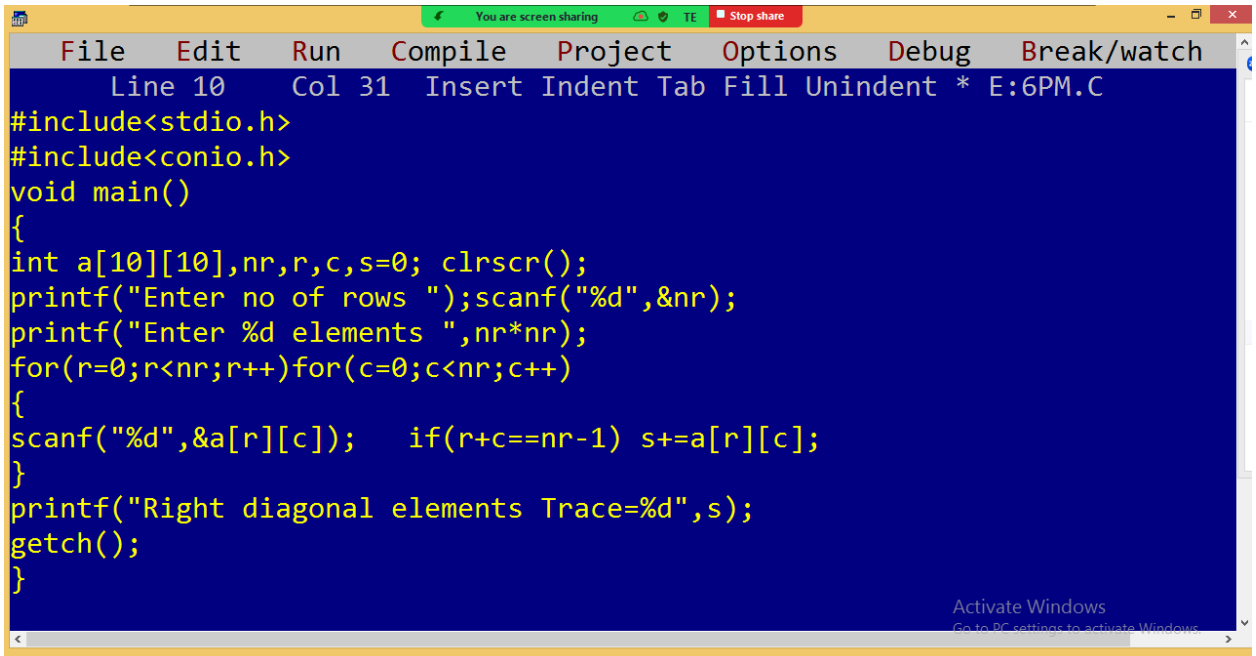
r: 0 1 2

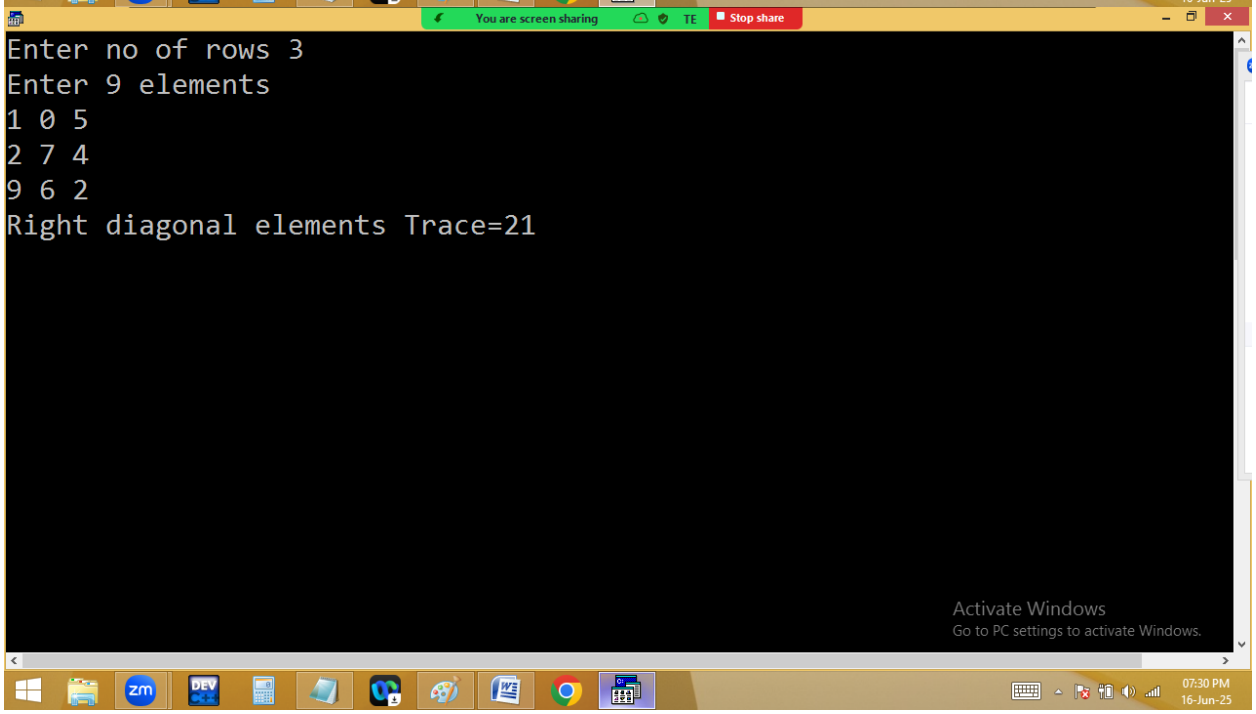
c: 0 1 2
0 1 2
0 1 2

s: 0+1+7+2 = 10

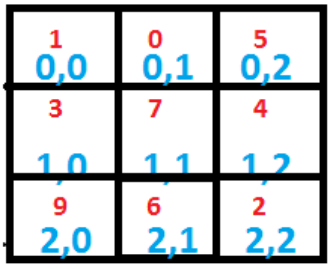
Trace ==> 1+7+2=10

Find the trace of right diagonal elements:


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 31 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,r,c,s=0; clrscr();
printf("Enter no of rows ");scanf("%d",&nr);
printf("Enter %d elements ",nr*nr);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)
{
scanf("%d",&a[r][c]); if(r+c==nr-1) s+=a[r][c];
}
printf("Right diagonal elements Trace=%d",s);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:30 PM
16-Jun-25


You are screen sharing
Stop share
Enter no of rows 3
Enter 9 elements
1 0 5
2 7 4
9 6 2
Right diagonal elements Trace=21
Activate Windows
Go to PC settings to activate Windows.
07:30 PM
16-Jun-25

Home work:



	Even	Odd	Zero
1-row	0	2	1
2-row	1	2	0
3-row	2	1	0